5.1 Machin's Formula

WORKSHEET

The question

(i) If $z = x + \mathrm{i}y$, with x, y real, show that

$$|x|\cos\alpha + |y|\sin\alpha \leq |z| \leq |x| + |y|$$

for all real α.

(ii) By considering $(5 - \mathrm{i})^4(1 + \mathrm{i})$, show that

$$\frac{\pi}{4} = 4\tan^{-1}\left(\frac{1}{5}\right) - \tan^{-1}\left(\frac{1}{239}\right).$$

Prove similarly that

$$\frac{\pi}{4} = 3\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{1}{20}\right) + \tan^{-1}\left(\frac{1}{1985}\right).$$

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself What is the argument of $5 - \mathrm{i}$? Of $1 + \mathrm{i}$? What happens to arguments when complex numbers are multiplied?

STAGE 1 Bounds on the modulus

Ask yourself For the upper bound, square both sides. For the lower bound, think of $|x|\cos\alpha + |y|\sin\alpha$ as a dot product of two vectors.

STAGE 2 The key multiplication

Ask yourself Compute $(5 - \mathrm{i})^2$, then square it, then multiply by $1 + \mathrm{i}$.

Pause and try Now compare arguments. What is the argument of $956 - 4\mathrm{i}$? Why are you sure there is no extra multiple of 2π?

STAGE 3 The second formula

Ask yourself By analogy, which product should you compute this time?

Try this next

1. **A cousin.** (From STEP II, 2000.) By considering $(5 - \mathrm{i})^2(1 + \mathrm{i})$, prove that $\tan^{-1}\frac{7}{17} + 2\tan^{-1}\frac{1}{5} = \frac{\pi}{4}$.
2. **Euler's formula.** Show that $\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3} = \frac{\pi}{4}$ by considering $(2 + \mathrm{i})(3 + \mathrm{i})$.
3. **Hutton's formula.** Show that $\frac{\pi}{4} = 2\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{7}$.
4. **How many terms?** Estimate how many terms of each series in Machin's formula you would need to compute π to 20 decimal places. Then do it for the first few digits with a calculator.

Further reading

- **Jonathan Borwein & Peter Borwein, *Pi and the AGM*** (Wiley). Includes the history of Machin-type formulas.

- **Jörg Arndt & Christoph Haenel, *Pi Unleashed*** (Springer). A lively account of methods for computing π, old and new.
- **Tristan Needham, *Visual Complex Analysis*** (Oxford University Press). The best book for seeing complex numbers geometrically. Chapter 1 is about multiplication as rotation and scaling.

The full guided solution, and what the question leads to at university, is in Section 5.1:
https://beyondstep.keithmcnulty.org/ch05/machin.html